
```
clc;
clear;
close all;
```

Parameters

```
N = 100;           % Number of bits
sps = 20;          % Samples per symbol
```

Generate Random Binary Data

```
data = randi([0 1], 1, N);
```

NRZ Encoding

```
tx = repelem(data, sps);
```

Time Axis

```
t = (0:length(tx)-1) / sps;
```

Introduce Inter-Symbol Interference (ISI)

Simple Low-Pass FIR Channel

```
h = [0.2 0.6 1 0.6 0.2];
```

Received Signal with ISI

```
rx = conv(tx, h, 'same');
```

Plot Results

```
figure('Name', 'Inter-Symbol Interference (ISI)', ...
       'NumberTitle', 'off');
```

```
% Original Binary Data
subplot(4,1,1);
stairs(data, 'LineWidth', 1.5);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
ylabel('Bit');
ylim([-0.2 1.2]);
```

```
% Transmitted NRZ Signal
subplot(4,1,2);
```

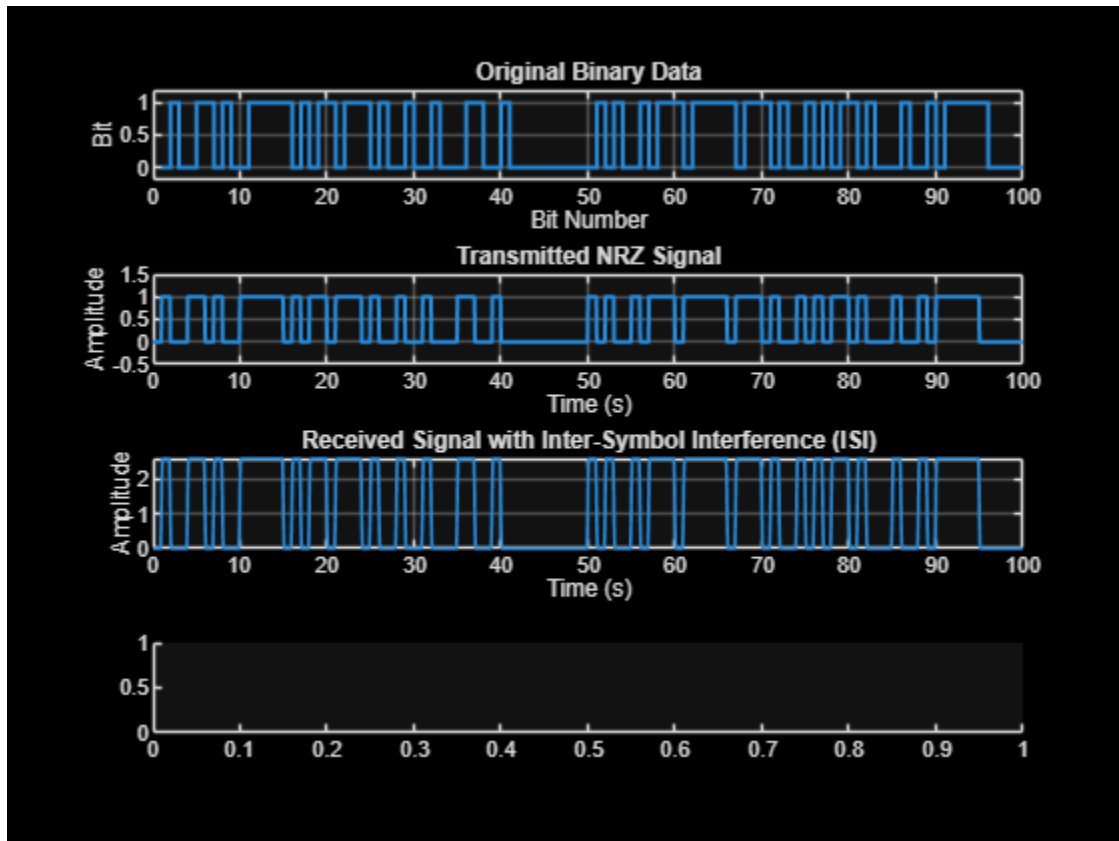
```

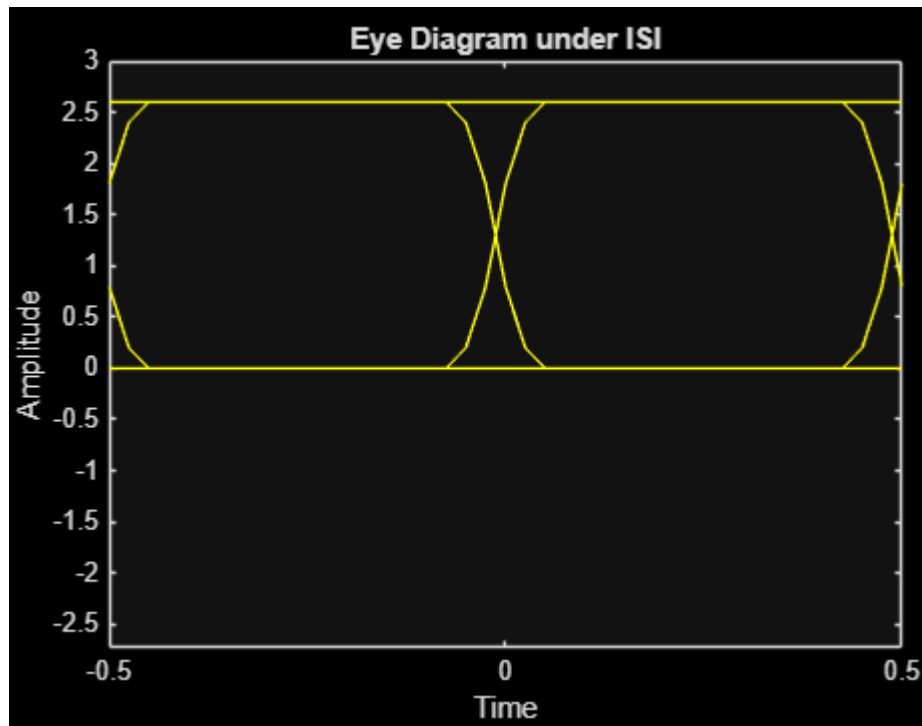
plot(t, tx, 'LineWidth', 1.5);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time (s)');
ylabel('Amplitude');
ylim([-0.5 1.5]);

% Received Signal with ISI
subplot(4,1,3);
plot(t, rx, 'LineWidth', 1.2);
grid on;
title('Received Signal with Inter-Symbol Interference (ISI)');
xlabel('Time (s)');
ylabel('Amplitude');

% Eye Diagram
subplot(4,1,4);
eyediagram(rx, 2*sps);
title('Eye Diagram under ISI');

```





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